

Lab 4

Class / Object / Constructor in OOP (Java - NetBeans)

Objective:

- To understand the concept of **classes and objects** in Java.
- To learn how **constructors** are used to initialize objects.
- To practice using **methods, mutators, and accessors** in programming problems.

Theory:

In Java, a class is a blueprint used to create objects. It defines the properties (attributes) and behaviors (methods) of an object. An object is an instance of a class that represents real-world entities.

A constructor is a special method that is automatically called when an object is created. It is mainly used to initialize the attributes of the object. Java supports different types of constructors such as default constructors and parameterized constructors, and it also allows constructor overloading.

In addition, mutator methods (setters) are used to assign values to variables, while accessor methods (getters) are used to retrieve values. These concepts help in organizing and managing data effectively in object-oriented programming.

Syntax:

```
class ClassName {  
    // Attributes (Variables)  
    dataType variable1;  
    dataType variable2;  
    // Default Constructor  
    ClassName() {  
        // initialization code  
    }  
    // Parameterized Constructor  
    ClassName(dataType param1, dataType param2) {  
        // initialization code  
    }  
    // Mutator (Setter)  
    void setVariable(dataType value) {  
        variable1 = value;  
    }  
    // Accessor (Getter)  
    dataType getVariable() {
```

```
        return variable1;
    }
    // Method
    returnType methodName() {
        // code
    }
}
// Creating Object
ClassName obj = new ClassName();
```

Task 1:

Write a class named “Addition” that has the following member variables:

1. Num1: integer variable
2. Num2: integer variable
3. Res: integer variable

The class should have the following functions:

1. Mutators: to set the values for num1 and num2.
2. Accessors: to return the values for num1, num2 and res.
3. Default constructor assigns the value 0 to all member variables.
4. Add: to add num1 and num2 and return res as the summation result.

Demonstrate the class in a program that:

1. Creates an object named A1 with value 5 for num1 and value 6 for num2.)
2. Create another object named A2
3. Call the add function for both object A1 and A2.

Object Diagram:

A1 → num1=5, num2=6, res=11

A2 → num1=0, num2=0, res=0

Task 2: Write a JAVA program for this class diagram

Person
- FirstName: string - LastName: string - FullName: string
+ Person() + setFirstName(fname) + setLastName(lname) + getFirstName() + getLastName() + getFullName() + ConcatenateName()

Demonstrate the class in a program that:

1. Creates an object named P1 with the FirstName as "John" and LastName as "Doe".
2. Creates another object named P2.
3. Call the ConcatenateName function for both objects P1 and P2.

Task 3:

Book

- title: string
- author: string
- publicationYear: int

- + Book ()
- + Book (string, string, int)
- + Book (Book&)
- + ~Book ()
- + getTitle (): string
- + getAuthor (): string
- + getPublicationYear (): int
- + setTitle (string)
- + setAuthor (string)
- + setPublicationYear (int)
- + DisplayBookInfo ()

Demonstrate the class in a program that:

1. Create an object named "book1" using the parameterized constructor (title = "To Kill a Mockingbird," author = "Harper Lee" publication year = 1960, page count = 281). Display the book's information using the "DisplayBookInfo" function.

2. Create an object named "book2" using the default constructor. Set the title to "The Great Gatsby," the author to "F. Scott Fitzgerald" and the publication year to 1925 using mutator functions. Display the book's information.